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When Does SHAP Outperform LIME? Model- and Configuration-Dependent Results from a Paired Tabular Benchmark

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Abstract:

Choosing between post-hoc explainers for a tabular model is a decision about faithfulness, stability and computational cost, yet controlled comparisons under matched conditions remain scarce. This paper reports a paired benchmark of SHAP, LIME, Anchors and DiCE in which every explainer sees the same model, the same test instances and the same seed, on one recorded host. The primary benchmark covers five model families on the Adult census-income dataset; six further datasets, chosen to cross feature dimension with feature type, and a configuration sweep are reported as exploratory evidence. Cost is reported as CPU time and as the number of model evaluations, which does not depend on the hardware. [R: main result for fidelity and stability, with effect sizes.] [R: result for Anchors and DiCE.] [R: cost result, including whether the TreeExplainer ordering between XGBoost and random forests holds when both ensembles are matched in size.] [R: whether the stability ordering holds over the LIME kernel width and the KernelSHAP budget.] The results support choosing the explainer by model architecture, configuration and latency budget, and they state which claims hold beyond the configurations tested.

Keywords: *Benchmarking, Computational cost, Explainable artificial intelligence, Explanation evaluation, Feature attribution*

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1. INTRODUCTION

A team that deploys a tabular classifier and must explain its predictions faces a concrete decision: which post-hoc explainer to use, for which model, under which latency budget. The candidates differ in the kind of explanation they return. LIME [1] and SHAP [2] return feature attributions, Anchors [3] returns a rule and DiCE [4] a counterfactual. They also differ in how faithfully the explanation tracks the model, how stable it is under small changes of the input, and how much computation it needs [5–7]. Theory does not settle the choice across model families and configurations [8, 9], and the empirical guidance is weaker than the popularity of the methods suggests: evaluations use different metric families, and the ranking of explainers often depends as much on the metric and the configuration as on the methods [10–12].

Two gaps follow. Comparisons in which all explainers see the same model, the same instances and the same seed, so that a difference can be attributed to the explainer, are scarce: published SHAP–LIME comparisons are often case studies in a single domain [13, 14]. Results are also usually reported for one configuration, although the hyperparameters that build LIME's neighborhood [15] or bound KernelSHAP's sampling [7] can change the outcome. Cost, finally, transfers to another deployment only when it is measured on a stated host, as item 2c of the REFORMS checklist for machine-learning-based science recommends [16], and when it is also counted in a unit that does not depend on that host, such as the number of model calls.

This paper addresses these gaps with three contributions:

1. **A paired benchmark of four explainers** (SHAP, LIME, Anchors and DiCE) on five model families, with every run on one recorded host, and cost reported as CPU time and as the number of model evaluations, a measure that does not depend on the hardware (Sections 4.1 and 4.2).
2. **Configuration-aware results**: a sweep over the LIME kernel width and sample size and over the KernelSHAP budget and background, and controls that test whether the stability measurement itself is valid (Sections 4.3 and 4.4).
3. **Bounded recommendations**: every claim is labelled as confirmatory or exploratory and stated with the models, datasets and configurations under which it holds (Section 5).

The paper makes ranking claims under a stated protocol. It does not claim that an explanation is correct for a task, and it does not test whether people decide better with any explainer.

2. BACKGROUND AND RELATED WORK

LIME fits an interpretable surrogate g to the model f on perturbed samples around an instance x, weighted by a proximity kernel π_x, as in Eq. (1):

$$\xi(x) = \operatorname{argmin}_{g \in G} \mathcal{L}(f, g, \pi_x) + \Omega(g), \tag{1}$$

where G is the class of surrogates (sparse linear models here), $\mathcal{L}$ measures how poorly g reproduces f on the weighted samples and $\Omega(g)$ penalizes surrogate complexity [1]. The kernel width, the number of samples and the feature budget shape the neighborhood, and the sampling makes the explanation stochastic. SHAP assigns additive attributions $f(x) = \phi_0 + \sum_{i=1}^{M} \phi_i$ that satisfy local accuracy, missingness and consistency, where ϕ_0 is the expected output over a background sample and M the number of features [2]. KernelSHAP estimates them by a weighted regression over sampled feature coalitions and applies to any model; TreeSHAP computes them from the structure of tree ensembles. Anchors searches for a rule that fixes the prediction with a target precision [3]; DiCE searches for a counterfactual instance with the opposite prediction [4].

Quantitative evaluation relies on proxy metrics: faithfulness under feature removal [17, 18], robustness to small input changes [19] and sparsity, standardized in toolkits and benchmarks [20, 21]. Removal-based scores depend on the removal scheme [22], attributions can pass perturbation tests and fail sanity checks [23], LIME and SHAP can be manipulated [24], and Shapley attributions are hard to read as importance when features are dependent [25]; feature attribution has provable limits as a basis for inferring model behavior [26], and attribution methods can mark features with no relation to the target as important [27].

Recent work compares SHAP and LIME in applied settings, among them education [13], software engineering [14] and mixed-type tabular models [28], and across tabular and image classifiers [6]; their respective strengths are reviewed in [29, 30]. Guidelines and open benchmarks for functionally grounded evaluation on tabular data are emerging [5, 31]. Estimators of Shapley values differ in accuracy and cost [7], LIME's neighborhood sampling can be redesigned to raise its fidelity [15], and counterfactual explainers can disagree on the same instance [32]. This study adds a comparison in which the four explainers are paired on the same instances, their configuration is varied, the validity of the stability measurement is tested, and cost is counted in model evaluations as well as in time.

3. MATERIALS AND METHODS

3.1. Design and Research Questions

All results come from one experimental cohort run for this study under the protocol below (Table 1). An earlier analysis of the SHAP–LIME question used runs collected on several hosts; an audit found that in most of its cells the evaluation instances overlapped the training partition, that one model could not be recovered and that its timings mixed hosts, so no number from it is used here.

For each endpoint Y and each instance, the paired difference is $\Delta_X = Y_{\mathrm{SHAP}} - Y_X$ for $X \in \{\mathrm{LIME}, \mathrm{Anchors}, \mathrm{DiCE}\}$. Three hypotheses state an expected direction; each is tested two-sided, and a difference is read in the stated direction only when its interval excludes zero on that side. Sparsity is reported descriptively:

- **H1 (stability).** SHAP is more stable than X: $\Delta_X(S) > 0$.
- **H2 (cost).** SHAP is more expensive than X: $\Delta_X(C) > 0$, tested in CPU time; the rows passed to the model are reported alongside, without a test.
- **H3 (faithfulness).** SHAP is more faithful than X: $\Delta_X(F) > 0$ and $\Delta_X(G) > 0$.
- **Sparsity (descriptive).** With its 10-feature budget LIME's sparsity is set by the user, and Anchors returns few features by construction, so a test of sparsity would test a configuration, not a method.

Table 1. Design of the benchmark. The primary benchmark is confirmatory; the other parts are exploratory

Factor	Levels
Datasets	Adult (primary, 108 encoded features); Breast Cancer (30), German Credit (61), Diabetes (8), Spambase (57), Phishing (68), Connect-4 (126)
Models	logistic regression, random forest, XGBoost, SVM (RBF), MLP
Explainers	SHAP, LIME (full vector; 10-feature variant), Anchors, DiCE
Seeds	Adult: 5; other datasets: 3
Instances	up to 25 per stratum (TP, TN, FP, FN) of the test split, per model and seed; stability on 3 per stratum
Paired cells	Adult: 25 (model × seed); other datasets: 90
Timing subset	5 instances per stratum, 3 repetitions, one host
Sweep and controls	Section 4.3 and Section 4.4

3.2. Data and Models

Adult [33, 34] is the primary dataset. Six further binary datasets were chosen so that feature dimension crosses feature type: Breast Cancer [35] and German Credit [36] from the UCI repository [37], and Diabetes, Spambase, Phishing and Connect-4 [38–41] in a 2×2 of numeric or categorical features and lower or higher dimension. For Connect-4, a win is the positive class and the two other outcomes the negative class.

Each dataset has one fixed stratified split (80/20, seed 42), shared by preprocessing, model training and every seed. Numeric features are scaled and categorical features one-hot encoded, both fitted on the training split only. Training is capped at 20,000 rows (a stratified subsample) so that the SVM fits in reasonable time. The five models [42, 43] are trained once per dataset with fixed hyperparameters (supplementary file) and frozen; every run checks the model file against its recorded hash. A seed therefore changes the evaluation instances and the explainers' randomness, never the model. Evaluation instances are drawn from the test split only, up to 25 from each confusion-matrix stratum; smaller strata are taken whole, and the realized counts are reported. Within a cell, all explainers see the same instances. In small strata, different seeds can draw the same instance; the overlap is measured and reported [R: overlap].

3.3. Explainers

All explainers receive the same encoded features and explain the predicted probability of the positive class:

- **SHAP:** TreeSHAP for the random forest and XGBoost; KernelSHAP for the other models, with 50 background rows and a budget of 500 coalitions.
- **LIME:** 1,000 samples and the library's default kernel width, $0.75\sqrt{M}$ (≈ 7.79 on Adult). The primary contrast uses the full attribution vector; a variant with the 10-feature budget shows the effect of the budget.
- **Anchors:** precision target 0.95; the attribution vector marks the features of the anchor with 1 and the others with 0.
- **DiCE:** one counterfactual of the opposite class (random search); the attribution of feature j is $|x_j - x'_j|$.

Anchors (a rule) and DiCE (a counterfactual) enter as reference explainers of other kinds: the research questions remain about SHAP and LIME, and the two references show whether a SHAP–LIME difference is specific to the pair or shared by explainers in general. These settings are the libraries' defaults where defaults exist, so that a practitioner's first run is what is measured. KernelSHAP's budget is set explicitly because the default ($2M + 2048$ coalitions) made the SVM the cost driver of the whole design; Section 4.3 measures the effect of the default against the explicit budget. Anchors and DiCE return explanations of different kinds, so their vectors measure which features the explanation uses, not signed contributions; the metrics below are comparable across the four methods in that sense.

3.4. Metrics

Let $f(x)$ be the predicted probability of the positive class and $e(x) \in \mathbb{R}^M$ the attribution vector; masking feature j replaces it with its training mean $\bar{x}_j$. The metrics follow common definitions [17–19]:

1. **Stability** S: the mean cosine similarity over all pairs of $T = 15$ explanations of perturbed copies $x + \varepsilon^{(a)}$, $\varepsilon^{(a)} \sim \mathcal{N}(0, \sigma^2 I)$, $\sigma = 0.1$, added to every encoded feature (scaled numeric and one-hot columns alike), with S computed on 3 instances per stratum, Eq. (2):

$$S = \frac{2}{T(T-1)} \sum_{1 \le a < b \le T} \cos\left(e^{(a)}, e^{(b)}\right). \quad (2)$$

- The cosine with an all-zero vector is set to 0; such vectors are counted and S is also reported without them. Noise on one-hot columns leaves the data manifold; Section 4.4 repeats S with category-preserving perturbations and without noise.

2. **Fidelity** F: the Pearson correlation, across the M features, between $|e_j(x)|$ and the prediction change $|f(x) - f(\tilde{x}^{(j)})|$ when feature j alone is masked, Eq. (3):

$$F = \rho\left(\{|e_j(x)|\}_{j=1}^{M},\ \{|f(x) - f(\tilde{x}^{(j)})|\}_{j=1}^{M}\right). \quad (3)$$

- Absolute values are used because the masked change is unsigned. When either vector is constant, F is undefined and set to 0; these cases are counted.

3. **Faithfulness gap** G: the prediction change when the $k = 5$ features with the largest $|e_j(x)|$ are masked together.
4. **Sparsity** P: the share of features with $|e_j(x)| > 10^{-4}$ (lower is sparser).
5. **Cost** C: CPU time per explanation and the number of rows the explainer passes to the model (Section 3.6).

3.5. Statistical Analysis

Instances are nested in seeds and seeds in models, so instance-level differences are not independent. For each endpoint and contrast, the instance-level differences Δ_{gsi} of model g, seed s and instance i are fitted by a linear mixed model $\Delta_{gsi} = \mu + u_g + v_{gs} + \epsilon_{gsi}$, with random intercepts for the model and for the seed within the model [44]; μ is the effect, reported with its 95% confidence interval and two-sided Wald p-value; if the model cannot be fitted, a t-test on the cell means is used and the case is reported. Because five models are few clusters, the interval is checked by a cluster bootstrap that resamples models [45]. Only instances valid for both explainers of a contrast enter it. On the cell-mean differences, the standardized paired difference d_z [46], the Hodges–Lehmann estimate [47] and the matched-pairs rank-biserial correlation [48], the effect size of the Wilcoxon signed-rank test [49], are reported. The confirmatory family is four endpoints (S, F, G and CPU time) times three contrasts; p-values are Holm-corrected over these twelve tests [50]. The four explainers are ranked on the cell means of S, F and G by the Friedman test with the Nemenyi critical difference [51]. The sweep, the controls and the six further datasets are exploratory and are reported without tests.

3.6. Cost Measurement and Computing Environment

All runs used one laptop (AMD Ryzen 5 4600H, 6 cores, 31.4 GB RAM, Windows 11), Python 3.11 with scikit-learn 1.7.1, XGBoost 3.1.2, SHAP 0.50.0, LIME 0.2.0.1, alibi 0.9.6 (Anchors) and dice-ml 0.12. Quality was computed in parallel worker processes; quality does not depend on speed. Cost was measured in a separate pass on a subset (Table 1): one process with one thread, pinned to one core, on mains power; the explainers interleaved per instance, after one warm-up explanation each; three repetitions, of which the median is used; and measurements paused while any other process used more than 10% of a core or the machine was in use. A wrapper counts every row the explainer passes to the model. Rows evaluated, unlike time, transfer to other hardware: multiplied by the cost of one model evaluation they give the explainer's cost on any host. For logistic regression and the MLP, whose cost per row is known, this is also given in floating-point operations.

4. RESULTS

4.1. Primary Benchmark on Adult

Table 2 reports the effects on the 25 Adult cells. [R: per contrast, the direction and size of the stability and faithfulness effects, and in how many cells the sign holds.] [R: Friedman ranking of the four explainers per endpoint.] [R: share of all-zero vectors and of undefined fidelity per explainer, and whether excluding them changes any effect.] The distribution of the instance-level differences is shown in the supplementary file.

Table 2. Primary benchmark (Adult, 25 cells). Per-method mean levels and paired effects Δ_X = SHAP − X: mixed-model estimate with 95% CI, Holm-adjusted p, standardized paired difference d_z, Hodges–Lehmann estimate (HL) and matched rank-biserial correlation r_{rb}. Cost: CPU ms per explanation

Endpoint	X	SHAP	X	Δ_X [95% CI]	p_{Holm}	d_z	HL	r_{rb}
Stability	LIME	[R]	[R]	[R]	[R]	[R]	[R]	[R]
	Anchors		[R]	[R]	[R]	[R]	[R]	[R]
	DiCE		[R]	[R]	[R]	[R]	[R]	[R]
Fidelity	LIME	[R]	[R]	[R]	[R]	[R]	[R]	[R]
	Anchors		[R]	[R]	[R]	[R]	[R]	[R]
	DiCE		[R]	[R]	[R]	[R]	[R]	[R]
Faithf. gap	LIME	[R]	[R]	[R]	[R]	[R]	[R]	[R]
	Anchors		[R]	[R]	[R]	[R]	[R]	[R]
	DiCE		[R]	[R]	[R]	[R]	[R]	[R]
Cost	LIME	[R]	[R]	[R]	[R]	[R]	[R]	[R]
	Anchors		[R]	[R]	[R]	[R]	[R]	[R]
	DiCE		[R]	[R]	[R]	[R]	[R]	[R]
Sparsity	all	[R]	[R]	descriptive				

4.2. Cost

Table 3 gives the cost per explanation by model family. [R: CPU-time ordering by model family; the share of model time in the explainer's time.] [R: rows evaluated per explanation: LIME 1,000; KernelSHAP about 25,000 (500 coalitions × 50 background rows); Anchors and DiCE as measured.] Earlier runs suggested that TreeSHAP was faster than LIME on XGBoost but slower on the random forest, two ensembles of different size. Here both have 100 trees, and the comparison is repeated with matched depth and with cost plotted against the total number of leaves [R: whether the reversal holds, and whether it follows the size of the forest or the implementation].

Table 3. Cost per explanation on Adult (one host, one core): median CPU time and rows evaluated by the model; theoretical complexity per explanation (T trees, L leaves, D depth; n_s samples or coalitions; n_b background rows). FLOPs for logistic regression and the MLP in the text

Explainer	Model	Complexity	CPU (ms)	Rows evaluated
SHAP (Tree)	RF, XGB	$O(TLD^2)$, no model calls	[R]	0
SHAP (Kernel)	logreg, SVM, MLP	$O(n_s n_b)$ rows	[R]	[R]
LIME	all	$O(n_s)$ rows + fit $O(n_s M^2)$	[R]	[R]
Anchors	all	adaptive (precision target)	[R]	[R]
DiCE	all	iterative search	[R]	[R]

4.3. Configuration Sensitivity

The sweep varied, on Adult for every model and three seeds, LIME's kernel width and number of samples, and KernelSHAP's budget and background size; KernelSHAP was also run on the two tree ensembles, to separate the method from the TreeSHAP implementation. The grid crosses LIME kernel widths of 0.75, 1.5, 3, 5, 10 and 15 and the default $0.75\sqrt{M}$ with 250 to 5,000 samples, KernelSHAP budgets of 100 to 1,000 coalitions and the default with 10 to 100 background rows, and TreeSHAP with 10 to 100 background rows; each cell has one instance per stratum, with stability on each. [R: Figure 1 placeholder: Pareto fronts of stability, fidelity and cost per model, one panel per model, computed on the instances valid under every configuration.] [R: whether the sign of each SHAP–LIME difference holds over the grid; the share of degenerate explanations at small kernel widths; the effect of the library default against the 500-coalition budget.]

4.4. Validity of the Measurements

Five controls test whether the stability and fidelity measurements measure the explainer: (a) explanations of the unperturbed instance repeated T times, which isolates each explainer's own sampling variance; (b) noise that keeps one-hot features valid (a category is resampled instead of adding Gaussian noise to its indicators); (c) rank correlation and top-k overlap on the full vectors, besides the cosine; (d) the effect of excluding all-zero vectors; and (e) the fidelity ordering under four masking schemes (zero, mean, marginal and grouped replacement, where associated features such as education and its numeric code are masked together) on every model. [R: for each control, whether the SHAP–LIME stability and fidelity orderings hold, and how much of LIME's instability is its own sampling variance.]

4.5. Other Datasets

Table 4 reports the four explainers on the six further datasets, now including SHAP stability. [R: whether the Adult orderings hold across the 2×2 of dimension and feature type; where they change.]

Table 4. Other datasets (exploratory): mean ± SD over 3 seeds, averaged over the five models (per model in the supplementary file)

	Stability		Fidelity					
Dataset	SHAP	LIME	Anch.	DiCE	SHAP	LIME	Anch.	DiCE
Breast Cancer	[R]	[R]	[R]	[R]	[R]	[R]	[R]	[R]
German Credit	[R]	[R]	[R]	[R]	[R]	[R]	[R]	[R]
Diabetes	[R]	[R]	[R]	[R]	[R]	[R]	[R]	[R]
Spambase	[R]	[R]	[R]	[R]	[R]	[R]	[R]	[R]
Phishing	[R]	[R]	[R]	[R]	[R]	[R]	[R]	[R]
Connect-4	[R]	[R]	[R]	[R]	[R]	[R]	[R]	[R]

5. DISCUSSION

5.1. What Holds, and Where

Table 5 states each claim with its evidence and its scope. Confirmatory claims rest on the primary benchmark and its corrected tests; exploratory claims rest on the sweep, the controls and the further datasets, and are reported as patterns, not as tested effects. [R: two or three sentences on the main pattern, including the conditions under which an ordering changes.]

Table 5. What holds, where, and under which configuration

Claim	Status	Scope
SHAP more stable than X	confirmatory	[R]
SHAP more faithful than X	confirmatory	[R]
SHAP more expensive than X	confirmatory	[R]
Orderings over the configuration grid	exploratory	[R]
Orderings on other datasets	exploratory	[R]
TreeSHAP vs LIME cost by ensemble	exploratory	[R]
Usefulness for people	not tested	no participant study

The recommendations are restricted to what was tested. [R: where SHAP is preferable on faithfulness and stability and at what cost; where a cheaper explainer suffices; which configuration choices a practitioner should check before relying on either rule.]

5.2. Relation to Earlier Work

Earlier comparisons of SHAP and LIME, including a multi-metric benchmark on Adult [52], reported SHAP ahead on fidelity and stability and LIME ahead on cost, for one configuration and with timings from unrecorded hardware. This study differs in four respects: all four explainers are paired on the same instances; every run uses one recorded host and cost is also counted in model evaluations; the configuration is varied and the measurement is tested; and the results extend to six further datasets. [R: which earlier statements are confirmed, qualified or reversed.]

5.3. Limitations

The endpoints compare explainers under a protocol; they do not establish that an explanation is correct, which needs ground truth or human-centered evaluation [53], and attribution methods can mark suppressor variables as important when features are dependent [27]. Anchors and DiCE vectors encode which features an explanation uses, not signed contributions. Cost was measured on one host; times scale with the hardware, rows evaluated do not. The study covers tabular binary classification with five model families and seven datasets.

6. CONCLUSIONS

[R: two to three sentences that restate only results of Table 2 and Table 5: the confirmatory orderings with effect sizes, the cost in rows evaluated, and the conditions under which an ordering changes.] The choice of explainer depends on the model architecture, the configuration and the latency budget, and a ranking obtained under one configuration should not be transferred to another without measuring it. Whether any of these explainers helps people decide is a separate question that needs a study with participants.

Competing Interests

No competing financial interests or personal relationships that could have appeared to influence the work reported in this paper are declared. No external funding was received for this study.

Data and Code Availability

The code, the frozen models with their hashes, every result row and the analysis scripts are available in a public repository and a versioned archive with a DOI, under the MIT License; their addresses are withheld for masked review and are given on the title page. A supplementary file lists the design, the

hyperparameters, the per-dataset and per-model results and the commands that regenerate every table from the result rows.

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